

OBJECT ORIENTED PROGRAMMING

Course Code: CS-112

Credit Hours: 4(3+1)

Pre-requisite: Programming Fundamentals

Course Objectives

The key objectives of this course include:

To understand, and implement Object Oriented paradigm through Java.

To gain the ability to understand and navigate through API docs and utilizing different java packages.

To implement various java utilities such as graphical user interface, exception handling and, multi-threading etc.

Acquire the skills to develop and run Java desktop applications.

Course Learning Outcomes

After successful completion of this course, the students should be able to:

	GA	BT Level
1. Understand the underlying concepts of object-oriented paradigm.	2	C2
2. Examine and comprehend the object-oriented design artifacts and their mapping to object-oriented programming.	3	C4
3. Apply object-oriented principles to design and execute solutions for small systems involving single/multiple objects.	4	C3

Course Contents

This course aims to provide the students with an introduction and general concepts about Java Programming(J2SE) and how to develop Java applications. Topics covered include the Java programming language syntax, Procedural vs. Object oriented Programming, Principles of Object-oriented Programming, Data Abstraction, Encapsulation, Classes and Objects, Operator Overloading, Inheritance and its types, Polymorphism, Java Packages and Abstract Classes, Exception handling, File input/output, Threads, Swing and Database connectivity. Students will develop and test Java applications (typically) using Netbeans/ IntelliJ Idea.

Mapping of CLOs to GA's

GA's/CLOs	CLO 1	CLO2	CLO3
GA1 (Academic Education)			
GA2 (Knowledge for Solving Computing Problems)	✓		
GA3 (Problem Analysis)		✓	
GA4 (Design/Development of Solutions)			✓
GA5 (Modern Tool Usage)			
GA6 (Individual and Teamwork)			
GA7 (Communication)			
GA8 (Computing Professionalism and Society)			
GA9 (Ethics)			
GA10 (Life-long Learning)			

Resources

Text Book:

1. Java: How to Program by Paul Deitel, 12th Edition, 2024

Reference:

1. Beginning Java 2, by Ivor Horton, 5th Edition, 2005
2. An Introduction to Object Oriented Programming with Java by C. Thomas Wu., 5th Edition , 2009

Weekly Course Plan

Week No	Topics Covered
WEEK 01	An Overview of Object Oriented Programming and Java • History of Java • Procedural Programming vs OOP
	• OOP principles (encapsulation, inheritance and polymorphism) • Simple java program • Introduction to Class diagram
WEEK 02	Fundamental Programming Structure in Java • Data Types • Variables • Operators • Lexical issues (white spaces, identifiers, literals, comments, separators and java keywords. • Operators (Arithmetic, relational, logical and bitwise operators) • Operators precedence • Constants
	Control Structures
WEEK 03	Introducing Classes • Classes and Objects • Initializing Class • Declaring Object and reference variables • Methods • Arguments and Parameters • Static Members
	• Anonymous Objects • Access specifiers • This pointer • Conversion of class diagram into code
WEEK 04	Constructors in Java • Default constructor • Parameterized constructor

	• Constructor overloading • Copy Constructor
	• Java finalize method • Function Overloading
WEEK 05	Operator Overloading • Operator Overloading methods • Restrictions on overloading • Overloading Unary Operator • Overloading Binary Operator
	Inheritance • Inheritance hierarchy • Fields and methods • Inheritance in UML diagram
WEEK 06	Inheritance • Composition • Super class and subclass, super class constructors, Super keyword • Types of Inheritance
	• Abstract methods and classes • Method overriding • Final keyword • Interfaces
WEEK 07	Polymorphism • Types of Polymorphism • Dynamic method dispatch
	• Java Packages • Methods to import java packages
WEEK 08	Dynamic Memory Allocation • Dynamic vs Static binding Java Packages • Java Core Packages • Java Sub-Packages
	• Practice Tasks
MID SEMESTER EXAMS	
WEEK 9&10	Arrays and Java Collections • Arrays, Declaring arrays, Array processing, Multidimensional arrays
	• Array Lists • Hash maps • Java Generics
WEEK 11	Introduction to GUI

	• Components • Containers
	• Simple GUI application. • J Components
WEEK 12	Layouts and Event Handling • Layouts • Look & Feel
	Event handling • Event Class • Event Listener Interfaces
WEEK 13	Exception Handling • Exception Handling fundamentals • Types of exceptions
	• Using exception clauses (try, catch, throw, throws, and finally) • Creating custom exceptions
WEEK 14	Multi-threaded Programming • Threads and process • Multi-threading fundamentals • Java thread model • The main thread • Thread states
	Threads Synchronization • The concurrency utilities • Thread synchronization
WEEK 15	File Handling & DB • Introduction to File Handling • File reading and Writing • Reading through Tokenizer • Introduction to DB
	Database Connectivity (with SQL Server) • JDBC • Query execution • Parameterized Query • Results sets
WEEK 16	Database Connectivity (with MS Access) • JDBC • Query execution
	• Parameterized Query • Results sets
END SEMESTER EXAMS	

OBJECT ORIENTED PROGRAMMING LAB

Course Code: CS-112L

LAB Objectives

The key objectives of this LAB include:

Understanding of Object-oriented concepts and basic functions and solving problems in JAVA.

To practice different JAVA language problems and implementing them in IntelliJ Idea.

To design and implement JAVA language programs for computational problems.

To test and debug programs written in JAVA language.

Lab Learning Outcomes

After successful completion of this course, the students should be able to:

	GA	BT Level
1. Practice key concepts of object-oriented programming	3	P3
2. Produce algorithms / codes for desired problems	4	P4
3. Express your solutions from problems conducted during your project and report.	6	A3

Lab Contents

Fundamental Programming Structure in JAVA, Introduction of classes, Inheritance, Polymorphism, Arrays and Java Collections, Introduction to GUI, Layouts and Event Handling, Event Handling, Exception Handling, Multi-Threaded Programming, Threads Synchronization, File Handling & DB, Database Connectivity (with SQL Server), Database Connectivity (with MS Access).

Mapping of LLOs to GA's

GA's/LLOs	LLO1	LLO2	LLO3
GA1 (Academic Education)			
GA2 (Knowledge for Solving Computing Problems)			
GA3 (Problem Analysis)	✓		
GA4 (Design/Development of Solutions)		✓	
GA5 (Modern Tool Usage)			
GA6 (Individual and Teamwork)			✓
GA7 (Communication)			
GA8 (Computing Professionalism and Society)			
GA9 (Ethics)			
GA10 (Life-long Learning)			

Resources

Text Books:

1. The Object-Oriented Thought Process by Matt .A, Wiesfield, 5th Edition, 2019

Reference:

Object Oriented Programming with Java, M.T.Somashekara, 1st Edition, 2017

Interactive Object-Oriented Programming in Java: Learn and Test Your Programming Skills, by Vascaran Sarcar, 2nd Edition, 2016

Weekly List of Experiments

Sr.	Topics	LLO, GA
1	Implementation of basic programs in Java using NetBeans after setting up and configuring JDK and NetBeans	LLO2, GA5
2	Implementation of the Fundamental Programming Structure in Java	LLO2, GA5
3	Implementation of the concept of Classes	LLO2, GA5
4	Implementation of the concept of Inheritance	LLO2, GA5
5	Implementation of the concept of Polymorphism	LLO2, GA5
6	Implementation of the concepts of Arrays and ArrayLists in Java	LLO2, GA5
7	Implementation of the codes for developing GUIs in Java.	LLO2, GA5
8	Lab Sessional	LLO1, GA3
9	Implementation of the concepts of Layouts and Event handling	LLO2, GA5
10	Open Ended Lab	LLO3, GA7
11	Implementation of the concepts of Exception Handling	LLO2, GA5
12	Implementation of the concepts of Multi-threaded Programming	LLO2, GA5
13	Implementation of the concepts of Threads Synchronization	LLO2, GA5
14	Implementation of File Handling & DB	LLO2, GA5
15	Implementation of Database Connectivity (with SQL Server)	LLO2, GA5
16	Lab Sessional	LLO3, GA7